

Indian Statistical Institute
M.Tech. (CS), Mid-Semester of Second Semester Examination, 2025-26
Database Management Systems

Full Marks: 30

Date: 25-02-2026

Time: 2 Hours

Answer any *three* of the following questions

$3 \times 10 = 30$

1. (a) Suppose the pair relations R_1 and R_2 , both having t number of tuples, share a common attribute A . Further assume that the ratio between the number of tuples returned by the operations $(R_1 \bowtie R_2)$ and $(R_1 \supset R_2)$ is 1:5 and $(R_2 \bowtie R_1)$ and $(R_2 \supset R_1)$ is 1:3. If the number of unique elements in A independently for R_1 and R_2 are exactly 10% of the total elements, then what will be the maximum number of tuples returned by the operation $R_1 \bowtie R_2$? Recall that $R_1 \bowtie R_2$ and $R_1 \supset R_2$ denote the semijoin and antijoin operations, respectively.

- (b) Prove or disprove the following statement:

“For two non-identical relations R_1 and R_2 , $(R_1 - R_2) \neq (R_2 - R_1)$ will always hold.”

6+4

2. Consider a Cricket League RRL that follows a single Round-Robin structure, in which every team plays every other team once. The final Champion and Runner-up are decided based on the maximum and second maximum number of wins, respectively. The following schema represents the details associated with the entire league. The attributes *team1*, *team2* and *team_winner* all take *team_id* values as entries. Note that the primary keys in the respective tables are underlined therein.

TEAMS = <team_id: integer, team_name: string, team_owner: string>

MATCH = <match_id: integer, team1: integer, team2: integer>

RESULT = <match_id: integer, team_winner: integer>

Write relational algebra expressions to return the following things.

- (i) Name(s) of the owner(s) who own multiple teams.
- (ii) Name(s) of the team(s) who never won a match in the RRL.
- (iii) Name(s) of the team(s) which became the Champion in the RRL.
- (iv) Name(s) of the team(s) which became the Runner-up in the RRL.

2+2+3+3

3. (a) Suppose a relation $R = \langle X, \underline{Y}, Z \rangle$ is to be indexed. Compare the performance of using a B^+ -tree and a hash table to execute the following operations in an effective way. Note that the underlined attribute is the primary key in the relation.

(i) $\sigma_{(Y > 0)}(R)$

(ii) $\sigma_{(X = 0) \vee (Z \neq 0)}(R)$

- (b) Consider the relations $R_1 = \langle X, Y, Z \rangle$ and $R_2 = \langle \underline{Y}, Z \rangle$ having 120 and 20 tuples,

respectively. Notably the primary keys in the respective relations are underlined therein. The number of distinct values of Y in R_1 and R_2 are the same. The range of values of X and Z in R_1 are $[100, 200]$ and $[-9, 10]$, respectively. Estimate the size of the following query considering the attribute values are uniformly distributed.

$$\sigma_{(X \leq 150) \vee (Z > 0)}(R_1) \bowtie \sigma_{(Y \neq 10)}(R_2)$$

(2+2)+6

4. Let there be a relation $R = \langle V, W, X, Y, Z \rangle$ having all the attributes as atomic and holding the functional dependencies $FD = \{V \rightarrow W, W \rightarrow X, X \rightarrow Y, Y \rightarrow W, VZ \rightarrow X\}$. Decompose R into subrelations (at least two) such that it satisfies all the following requirements.

(R1) The redundancy is eliminated from R as much as possible.

(R2) The decomposition is lossless join.

(R3) The decomposition is dependency preserving.

Show with explanations how the aforementioned requirements are satisfied in your proposed decomposition.

6+4